Problem-Solution Fit Canvas

Template: Problem-Solution Fit Canvas	**Maximum Marks: 5 Marks**
Date: 15 March 2026	**Team ID:** FV-AI-2026-X1
Project Name: FoodVerse AI	**Track:** AI/ML & Generative AI

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENTS [CS] • Health-conscious individuals & diabetics requiring strict macro tracking. • Urban households managing high grocery waste. • Commercial kitchens & restaurant hygiene inspectors.	**6. CUSTOMER LIMITATIONS [CL]** • Limited daily time to manually log food meals. • Standard smartphone cameras without specialized lab hardware. • Low tolerance for complicated, ad-heavy fitness apps.
2. PROBLEMS / PAINS + FREQUENCY [PR] • Guesswork in manual portion & calorie counting (Daily - 3x/day). • Silent produce decay & rot in refrigeration (Weekly). • Non-continuous subjective kitchen hygiene auditing (Continuous shift).	**5. AVAILABLE SOLUTIONS: PROS & CONS [AS]** • **MyFitnessPal:** Huge database, but manual and tedious. • **Visual inspection:** Free, but inaccurate for microbial decay. • **Manual paper audits:** Standard, but easily forged and sporadic.
3. TRIGGERS TO ACT [TR] • Doctor's diagnosis of pre-diabetes or hypertension. • Throwing away $40 worth of slimy, rotten produce at week's end. • Surprise health inspection warning or food safety violation notice.	**7. BEHAVIOR + INTENSITY [BE]** • Takes food photos before eating (High visual affinity). • Constantly googles 'Is this still safe to eat?' (High anxiety). • Skips manual logging when busy (High friction vulnerability).
4. EMOTIONS BEFORE / AFTER [EM] • **Before:** Anxious about dietary goals, guilty about household waste, stressed about commercial food safety. • **After:** In control with instant AI macro validation, confident about produce safety, reassured by compliance audit scores.	**8. CHANNELS OF BEHAVIOR [CH]** • **Online:** Mobile responsive browser on Vercel cloud, voice queries. • **Offline:** Kitchen countertop prep line, dining table, grocery pantry.
9. PROBLEM ROOT CAUSE [RC] Human sensory perception is incapable of estimating sub-surface biological decay indices and exact macronutrient densities without computational assistance.	**10. YOUR SOLUTION [SL]** **FoodVerse AI:** Unified multimodal intelligence combining Groq Qwen 3.8B vision reasoning, OpenCV color/texture analytics, and clinical metabolic dietetics into a 1-click cloud agent.